Name of the Experiment CALCULATION OF MEASURES OF DISPERSION

Date: 04.05.26
Exp. No.: 03
Page No.: 20

The following table gives the information of monthly expenditure of 130 farmers.

Expenditure (1000 tk)	10-20	20-30	30-40	40-50	50-60	60-70
No. of farmers	3	7	12	35	48	25

a) Calculate absolute measures of dispersion.

(i) Range (ii) Quantile deviation

(iii) Mean deviation (iv) Standard deviation

b) Calculate relative measures of dispersion

i) Co-efficient of range

ii) Co-efficient of quantile deviation

iii) Co-efficient of mean deviation

iv) Co-efficient of variation

Signature

Name of the Experiment CONTINUED

Date: 04.05.26

Exp. No.: 03

Page No.: 21

Table:

Expenditure (1000 tk)	Mid value (x_i)	Frequency (f_i)	Cumulative frequency	f_ix_i	$\lvert x_i-\bar{x}_2\rvert$	$f_i\lvert x_i-\bar{x}\rvert$	$(x_i-\bar{x})^2$	$f_i(x_i-\bar{x})^2$
10-20	15	3	3	45	34.85	104.55	1214.52	3643.36
20-30	25	7	10	175	24.85	173.95	617.52	4328.64
30-40	35	12	22	420	14.85	178.2	220.52	2646.24
40-50	45	35	57	1575	4.85	169.75	23.52	813.2
50-60	55	48	105	2640	5.15	247.2	26.52	1272.96
60-70	65	25	130	1625	15.15	378.45	229.52	5736
		$\Sigma f_i = N = 130$		$\Sigma f_ix_i = 6480$		$\Sigma f_i\lvert x_i-\bar{x}\rvert = 1252.4$		$\Sigma f_ix_i(x_i-\bar{x})^2 = 18466.6$

Solution:

Range = Highest value - lowest value

$= 70-10 = 60$

Here, Highest value $= 70$

lowest value $= 10$

$$\bar{x} = \frac{\Sigma f_ix_i}{N} = \frac{6480}{130} = 49.85$$

Signature

CamScanner

Name of the Experiment CONTINUED

Date: 04.05.26
Exp. No.: 03
Page No.: 22

Quantile deviation.

$$\frac{Q_3 - Q_1}{2}$$

$$Q_3 = L + \frac{3N/4 - Fm}{fm} \times h$$

$$= 50 + \frac{98-57}{48} \times 10$$

$$= 58.54$$

$$Q_1 = L + \frac{N/4 - Fm}{fm} \times h$$

$$= 50 + \frac{33-22}{35} \times 10$$

$$= 43.14$$

$3N/4 = \frac{3 \times 130}{4} = 97.5 \approx 98$

Both observation lies on (50-60)th class interval.

Here,
L = lower limit
Fm = Cumulative frequency of parameter class
fm = Frequency of median
N = No of observation
h = class interval

(ii) Quantile deviation $= \frac{58.54 - 43.4}{2}$

$= 7.2$

$Q_3 = 58.54$
$Q_1 = 43.14$

(iii) Mean Deviation $= \frac{1}{n} \Sigma f_i |x_i - \bar{x}|$

$= \frac{1}{130} \times 1254.4$

$= 9.63$

$n = 130$
$f_i |x_i - \bar{x}| = 1252.4$

(iv) Standard deviation $= \sqrt{\frac{1}{n} \Sigma f_i (x_i - \bar{x})^2}$

$= \sqrt{\frac{1}{130} \times 18466.6}$

$= \sqrt{141.83}$

$= 11.91$

here,
$n = 130$
$\Sigma f_i (x_i - \bar{x}) = 18466.6$

Signature

CamScanner

Name of the Experiment CALCULATION OF MEASURES OF DISPERSION

Date: 04.05.26

Exp. No.: 03

Page No.: 23

b) (i) Co.efficient of range $= \frac{X_{max} - X_{min}}{X_{max} + X_{min}} \times 100$

$= \frac{70-10}{70+10} \times 100$ | $X_{max} = 70$, $X_{min} = 10$

$= \frac{60}{80} \times 100 = 75\%$

(ii) Co.efficient of Quartile Deviation | $Q_3 = 58.54$, $Q_1 = 43.14$

$= \frac{Q_3 - Q_1}{Q_3 + Q_1} \times 100$

$= \frac{58.54 - 43.14}{58.54 + 43.14} \times 100 = \frac{15.4}{101.68} \times 100 = 15.15\%$

(iii) Co-efficient of mean deviation =

$= \frac{M.D(\bar{x}) \times 100}{\bar{x}}$

$= \frac{9.63}{49.85} \times 100 = 19.32\%$

(iv) Co-efficient of variation =

$= \frac{O\bar{x} \times 100}{\bar{x}}$

$= \frac{11.91}{49.85} \times 100$

$= 23.89\%$

12.5.26

$O\bar{x} = 11.91$

$\bar{x} = 49.85$

Signature

Name of the Experiment STUDY ON SHAPE CHARACTERS OF FREQUENCY DISTRIBUTION

Date: 11.06.26

Exp. No.: 04

Page No.: 24

Problem No-1: The following frequency distribution of height of 100 students in a class is as follows:-

Height(cm)	141-150	151-160	161-170	171-180	181-190
No. of students	5	16	56	19	4

Complete the first four central moments we slappard's connection for the 2nd and 4th central moments

Solution Table:

Class interval	Frequency (f_i)	Mid value (x_i)	$f_i x_i$	$x_i - \bar{x}$	$f_i(x_i-\bar{x})$	$f_i(x_i-\bar{x})^2$	$f_i(x_i-\bar{x})^3$	$f_i(x_i-\bar{x})^4$
141-150	5	145.5	727.5	-20.1	-100.5	2020.05	-40603.005	816120.4005
151-160	16	155.5	2488	-10.1	161.6	1632.16	-16484.886	166450.6416
161-170	56	165.5	9268	-0.1	-5.6	0.56	-0.056	5.6×10^{-3}
171-180	19	175.5	3334.5	9.9	188.1	1862.19	184315.681	1825132.2419
181-190	4	185.5	742	19.9	79.6	1584.04	31522.346	627295.6801
	$\Sigma f_i = 100$		$\Sigma f_i x_i = 16560$		$\Sigma f_i(x_i-\bar{x}) = 0$	$\Sigma f_i(x_i-\bar{x})^2 = 7098.98$	$\Sigma f_i(x_i-\bar{x})^3 = 7129.87$	$\Sigma f_i(x_i-\bar{x})^4 = 1792425.97$

$$\bar{x} = \frac{\Sigma f_i x_i}{N} = \frac{16560}{100} = 165.60$$

1st central moment $= \mu_1 = \frac{\Sigma f_i (x_i - \bar{x})}{N} = \frac{0}{100} = 0$

Here, $N = 100$

$\Sigma f_i (x_i - \bar{x}) = 0$

Signature

Name of the Experiment CONTINUED

Date: 11.06.26
Exp. No.: 04
Page No.: 25

2nd central moment $= \mu_2 = \frac{\sum f_i (x_i - \bar{x})^2}{N}$

$= \frac{7099}{100} = 70.99$

Here,
$\sum f_i (x_i - \bar{x}) = 7099$
$N = 100$

3rd central moment $= \mu_3 = \frac{\sum f_i (x_i - \bar{x})^3}{N}$

$= \frac{7129.8}{100}$

$= 71.29$

Here,
$\sum f_i (x_i - \bar{x})^3 = 7129.8$
$N = 100$

4th central moment $= \mu_4 = \frac{\sum f_i (x_i - \bar{x})^4}{N}$

$= \frac{1792425.92}{100}$

$= 17924.25$

Here,
$\sum f_i (x_i - \bar{x})^4 = 1792425.92$
$N = 100$

Shepard's connection for 2nd central moment

$\mu_2 (\text{connected}) = \mu_2 - \frac{h^2}{12}$

$= 70.99 - \frac{10^2}{12}$

$= 62.66$

Here,
$\mu_2 = 70.99$
$h =$ class interval
$\mu_4 = 17924.25$

Shepard connection for 4th central moment

$\mu_4 (\text{connected}) = \mu_4 - \frac{h^2}{2} \mu_2 + \frac{7}{240} h^4$

$= 17924.25 - \frac{10^2}{2} \times 70.99 + \frac{7}{240} \times 10^4$

$= 14666.42$

Signature

CamScanner

Name of the Experiment CONTINUED

Date: 11.06.26
Exp. No.: 04
Page No.: 26

Problem-02:

A distribution of short term credit distribuisement from 10 branches of a bank is given below:

Amount of credit (lakh)	0-5	5-10	10-15	15-20	20-25
No. of branches	1	2	4	2	1

Calculate first four central moments, coefficient of skewness and kurtosis and thus comment on the shape and nature of the distribution

Solution:

$$\text{Mean, } \bar{x} = \frac{\Sigma f_i x_i}{N}$$

$$= \frac{125}{10}$$

$$12.5$$

Here,
$N = 10$
$\Sigma f_i x_i = 125$

Table for solution is given at next page.

Signature

Name of the Experiment CONTINUED

Date: 11.06.26
Exp. No.: 04
Page No.: 27

Class interval	Freq^y (f_i)	Mid value (x_i)	$f_i x_i$	$x_i - \bar{x}$	$f_i(x_i-\bar{x})$	$f_i(x_i-\bar{x})^2$	$f_i(x_i-\bar{x})^3$	$f_i(x_i-\bar{x})^4$
0-5	1	2.5	2.5	-10	-10	100	-1000	10000
5-10	2	7.5	15	-5	-10	50	-125	1250
10-15	4	12.5	50	0	0	0	0	0
15-20	2	17.5	35	5	10	50	125	1250
20-25	1	22.5	22.15	10	10	100	1000	10000
	$\Sigma f_i = 10$		$\Sigma f_i x_i = 125$	$\Sigma x_i - \bar{x} = 0$	$\Sigma f_i(x_i-\bar{x}) = 0$	$\Sigma f_i(x_i-\bar{x})^2 = 300$	$\Sigma f_i(x_i-\bar{x})^3 = 0$	$\Sigma f_i(x_i-\bar{x})^4 = 22500$

1st central moment, $\mu_1 = \frac{\Sigma f_i(x_i-\bar{x})}{N}$

$= \frac{0}{10} = 0$

Here, $\Sigma f_i(x_i-\bar{x}) = 0$, $N = 10$

2nd central moment, $\mu_2 = \frac{\Sigma f_i(x_i-\bar{x})^2}{N}$

$= \frac{300}{10} = 30$

Here, $\Sigma f_i(x_i-\bar{x})^2 = 300$, $N = 10$

3rd central moment, $\mu_3 = \frac{\Sigma f_i(x_i-\bar{x})^3}{N}$

$= \frac{0}{10} = 0$

Here, $\Sigma f_i(x_i-\bar{x})^3 = 0$, $N = 10$

4th central moment, $\mu_4 = \frac{\Sigma f_i(x_i-\bar{x})^4}{N}$

$= \frac{22500}{10}$

$= 2250$

Here, $\Sigma f_i(x_i-\bar{x})^4 = 22500$, $N = 10$

Signature

CamScanner

Name of the Experiment: STUDY ON SHAPE CHARACTERS OF FREQUENCY DISTRIBUTION

Date: 11.06.26

Exp. No.: 04

Page No.: 28

Skewness $\beta_1 = \frac{\mu_3^2}{\mu_2^3} = \frac{0}{30^3} = 0$ | Here, $\mu_3 = 0$, $\mu_2 = 30$

Co efficient of skewness $\gamma_1 = \sqrt{\beta_1} = 0$

Comment: As, $\beta_1 = 0$, The distribution will be symetric

Kurtoss, $\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{2250}{(30)^2}$ | Here, $\mu_4 = 2250$, $\mu_2 = 30$

$= 2.5$

Co efficient of kurtosis, $\gamma_2 = \beta_2 - 3$ | Here $\beta_2 = 2.5$

$= 2.5 - 3$

$= -0.5$

Comment: As $\gamma_2 < 0$; the distribution will be platy kurtosis

Signature